

Premier University
Department of Computer Science and Engineering

Course Title: Machine Learning

Course Code: CSE 457

Assignment: 01 (CT-3)

Full Marks: 10

A fintech company in Bangladesh wants to build a system to detect fraudulent transactions.

Given Information:

- **Dataset size:** ~50,000 transactions
- **Features:**
 - Transaction amount
 - Time of transaction
 - User location
 - Device type
 - Transaction history
- **Target:** Fraud (Yes/No)
- **Problem type:** Binary classification
- **Challenges:**
 - Highly imbalanced data (fraud cases are rare)

Task-01: AI Based Suggestion (Mark-04)

Use an AI based tool (e.g., ChatGPT, Gemini) and ask for a solution.

Add:

- The Exact Prompt You Used
- AI's Full Response

Task 2: Critical Evaluation (6)

(Not more than 100 words)

- Critically Evaluate the AI's response
- Analyze strengths and weakness of AI's response
- Choose the most suitable model
- Draw an end to end system architecture (use draw.io / lucidchart)

Note:

→ Write down the 2nd Task of the assignment in your own words* (Not more than 100 words) and then draw a diagram.

→ *Submit a printed hardcopy with proper front page
also,*

→ ***Fill out the following google sheet***

https://docs.google.com/spreadsheets/d/1xE73v2Ua4M0wdbQS0N51akJ0ID9Jgmyvdc_rUd81NGA/edit?usp=sharing

**** I will check AI and similarities of the task 2 of your assignment.***